Identification of an Unknown - A

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Introduction

The goal of this experiment is to identify an unknown compound utilizing provided spectroscopy data from different spectroscopy methods. By interpreting the effects that the compound has on magnetic fields, light, and energy respectively, we can accurately estimate the identity of unknown compounds. It is necessary to understand the process of compound identification to ensure the safety of experimenters utilizing the compounds in experiments, while also hopefully maximizing the efficiency of the chemical reactions being tested.

Body

NMR

One such method of spectroscopic analysis is nuclear magnetic resonance (NMR). NMR analyzes an unknown compound by bombarding the molecules with strong, differential magnetic fields. These magnetic fields produce an electromagnetic, radio frequency (rf) signal that can affect the spin of the nuclei in the molecule and can then be further interpreted. This method is considered to be non-destructive, as there is no damage done to the molecules.

As the rf signals are interpreted, information about the change in energy between nuclei in the molecule can be calculated, and thus can give specific information regarding the location of atoms and functional groups in the molecule.

The NMR spectrometer itself can contain a very powerful magnet, liquid nitrogen, liquid helium, and an RF coil. The compound molecules in question are placed in the spectrometer and spun, thus creating a magnetic field. (Reusch, 2013)

FTIR

Another spectroscopic method used for unknown compound identification is Fourier transform infrared (FTIR) spectroscopy. FTIR utilizes an infrared spectrum of light to sweep over unknown compounds and examine the differing effects on the light caused by the specific nuclei of the compound. FTIR is especially helpful in determining the different functional groups within a molecule, as well as observing levels of emission or absorption of light across different phases of compounds. This method is also deemed as a non-destructive technique of identification. (2015).

The FTIR spectrometer sends infrared (ir) radiation at the molecule, measures how much of that radiation was absorbed or passed through the molecule, and outputs a chart that states the relationship between the wave numbers of the radiation waves and the percent transmittance of the specific functional groups in the compound.

GCMS

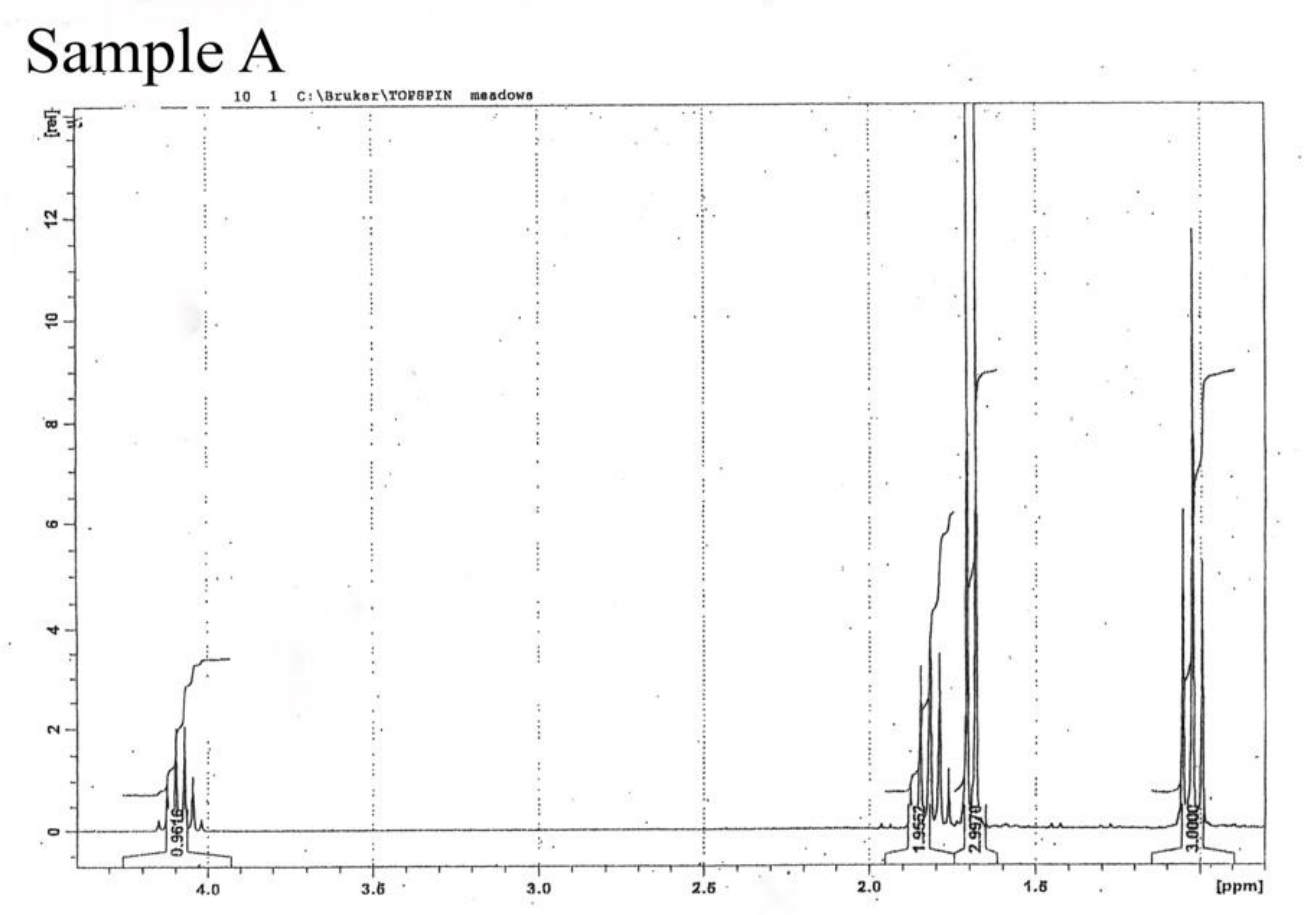
Finally, another effective spectroscopic method for identifying unknown compounds is gas chromatography mass spectroscopy (GCMS). This technique combines the use of a gas chromatograph to vaporize the unknown compound, and propulsion through an inert gas to separate the ions of the compound. Once the ions have been separated, natural properties of the atoms, such as boiling point, can be determined through the component's retention time, and thus the compound can be identified. (GC-MS)

This method, while very proficient in obtaining specific information about the compound, is usually preferred less than the previous methods because it is deemed as a destructive process. Separating the compound into its respective component causes damage to the molecule, and it is then unable to be reused in other experiments.

Data

The spectrometer data from each of the aforementioned methods of identification for the unknown compound "A" is shown below:

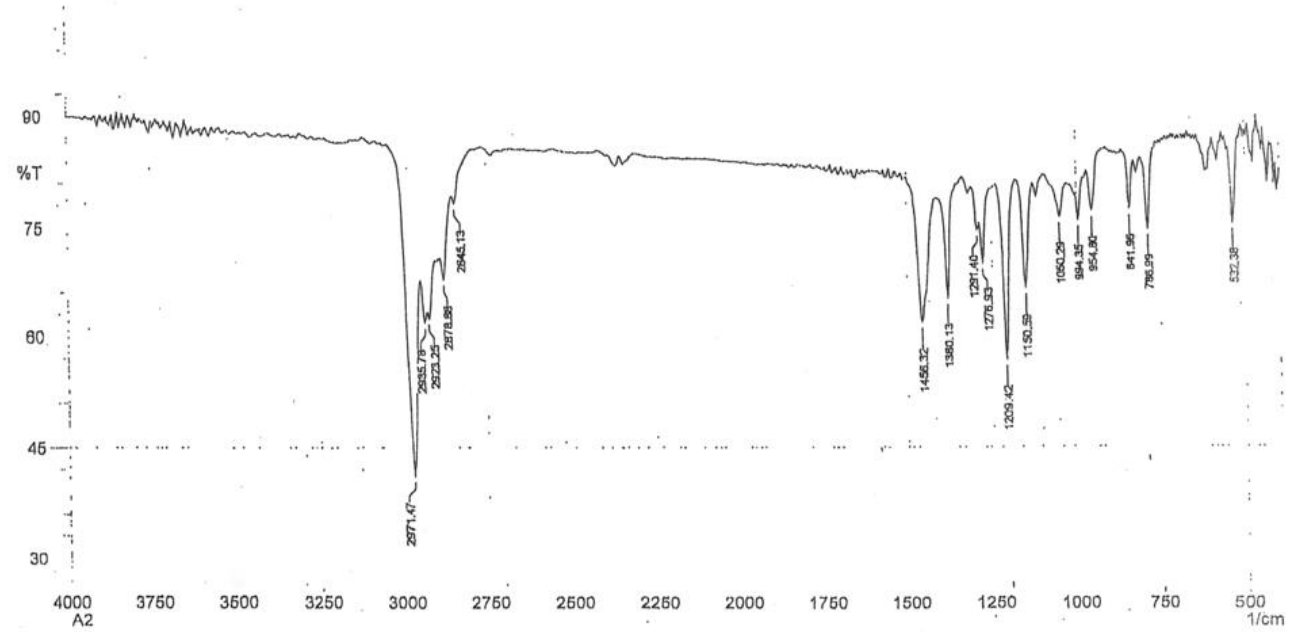
NMR Sample A



FTIR Sample A

SHIMADZU

Sample A

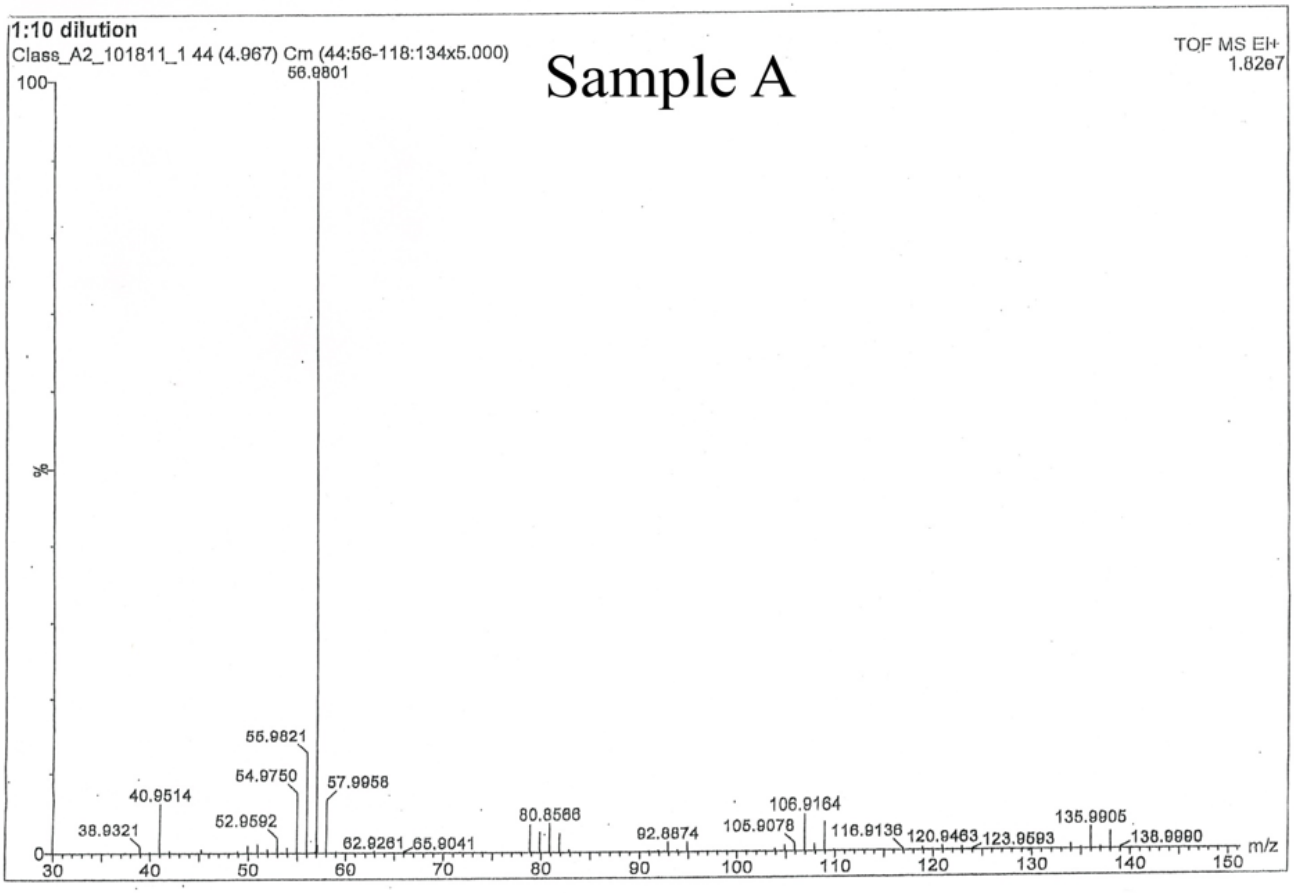


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GCMS Sample A



Analysis

NMR

Based on the data collected in the graph for the method of NMR spectroscopy, it is possible that the unknown compound “A” is C4H9Br. This approximation is majorly based on the functional groups and molecular structures observed in the graph.

Based on the analytical technique of integration values, reading the peak signals from left to right, the integration values are approximately 1, 2, 3 and 3. Because these are all whole numbers, no additional multiplication factor is necessary. This means that the first signal on the left represents 1 proton, the second represents 2 protons, the third represents 3 protons and the rightmost signal represents 3 protons. The left most signal could then be interpreted as CH. This signal seems to have 4 major peaks, which would indicate the presence of 3 neighboring protons, and the multiplicity is a quartet. The chemical shift for this signal is between 3.9 ppm and 4.5 ppm. Because this signal is at such a high chemical shift compared to the other signals, this indicates a high level of deshielding, which may be due to the presence of a halogen. The next two signals are somewhat close together. The left most of the two could be interpreted as a CH2. This signal seems to have 4 major peaks, which indicates 3 neighboring protons, and a quartet multiplicity. This signal is between a chemical shift of about 1.75 ppm and 1.9 ppm. The rightmost of the two could be interpreted as a CH3. This signal seems to have 2 major peaks, which indicates 1 neighboring proton, and a doublet multiplicity. This signal is between a chemical shift of about 1.6 ppm and 1.75 ppm. The rightmost signal could also be interpreted as a CH3. This signal seems to have 3 major peaks, which indicates 2 neighboring protons, and a multiplicity of triplet. This signal is between a chemical shift of about 0.5 ppm and 1.15 ppm.

FTIR

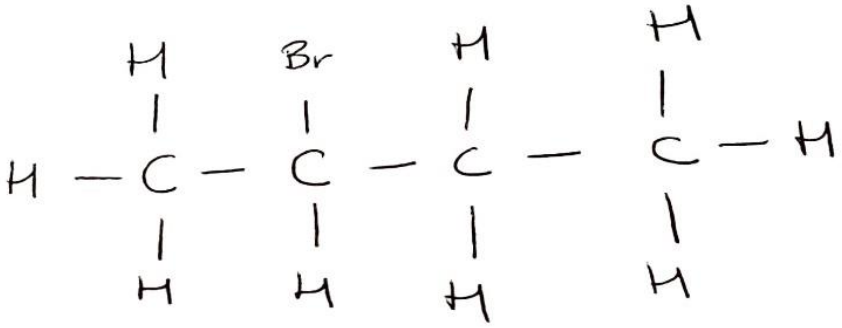
Based on the data collected in the graph for the method of FTIR spectroscopy, the approximation of the identity for unknown compound “A” from the NMR graph, C4H9Br, is confirmed. This confirmation is majorly based on the largest peaks outside of the fingerprint region, and their related wave numbers. The sharp point around the 3000-wave number region indicates the presence of an atom outside of C or H. However, since the region is fairly narrow, it is not likely to be a carbonyl, alcohol, O-H acid, or NH variation. It is more likely to be a halogen.

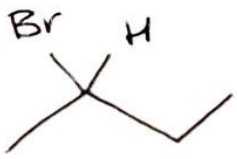
GCMS

Based on the data collected in the graph for the method of GCMS spectroscopy, the approximation of the identity for unknown compound “A” from the NMR graph and the FTIR graph is confirmed. This confirmation is majorly based on the functional groups and molecular structures observed in the graph. According to the GCMS graph, the most stable formation within the compound has an atomic mass of roughly 56.9501g. This is approximately the atomic mass of the C4H9 portion of the compound. The highest m/z value for atomic mass listed is approximately 138g. This is indicative of the highest mass of the entire compound. 138g-57g is approximately 81g. After searching for a halogen that had a similar molecular weight, it was determined that bromine was most likely the halogen in question for the compound.

Conclusion

After reviewing the data collected about unknown compound “A” across the three main techniques of spectroscopic identification, it is believed that compound “A” is C4H9Br, or 2-bromobutane.





References

FTIR analysis. RTI Laboratories: Environmental, Chemical & Materials Testing. (2015). https://rtilab.com/techniques/ftir-analysis/

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Reusch, W. (2013, May 5). Nuclear Magnetic Resonance Spectroscopy. NMR spectroscopy. https://www2.chemistry.msu.edu/faculty/reusch/virttxtjml/spectrpy/nmr/nmr1.htm